

4.1.10 Income Growth Data

According to economic theory, willingness-to-pay (WTP) to avoid air-pollution related morbidity effects and premature mortality should grow as real income increases. BenMAP-CE allows users to adjust the WTP estimates to account for the growth in income over time. This adjustment is a combination of data on income growth and estimated income elasticity of demand, which measures the responsiveness of the quantity demanded of a good to the change in the income of the people demanding the good; this is distinct from elasticity of demand, which quantifies the change in demand for goods and services as a result of changes in price for those goods and services. This section describes how to load the data adjusting for income growth and how EPA developed these adjustment factors.

Fundamental Concept – Income Growth Adjustment

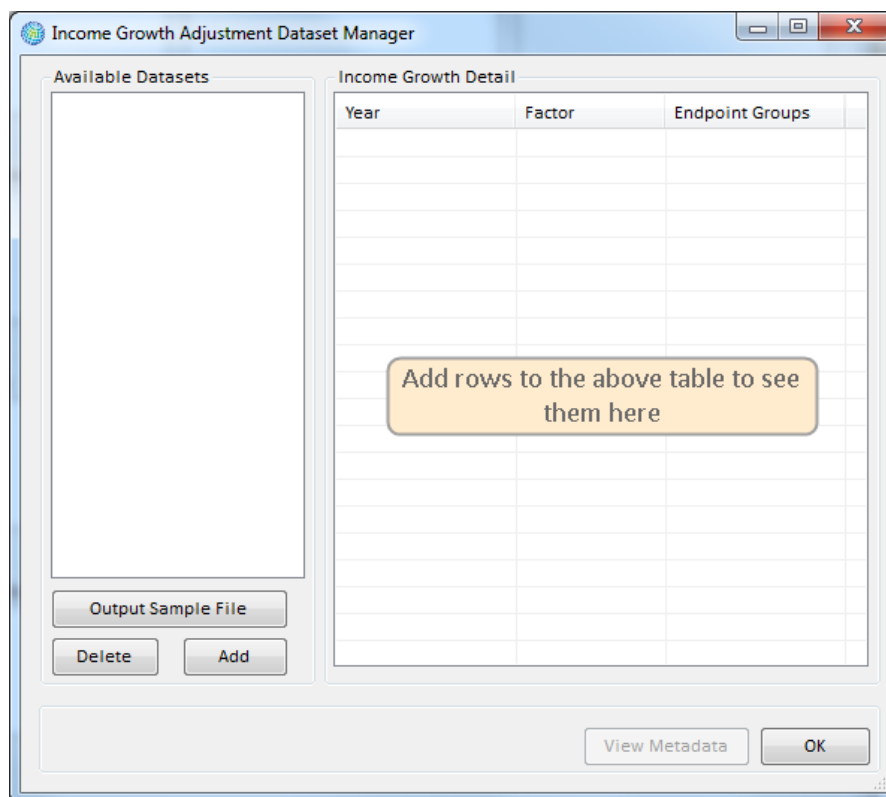
Income Growth Adjustment is the adjustment of the output of certain valuation functions to reflect the impact of increases in real income over time on WTP values. Generally, an increase in real income is expected to result in an increase in the willingness to pay (WTP). Users may select alternative values for the sensitivity of WTP to these changes.

Note that the WTP estimates in the default valuation functions in the *United States* setup are assumed to be based on 1990 income, so the income growth adjustments are all relative to 1990. That is, the income growth data has a value of 1 in 1990, and because income has generally increased over time in the U.S., the income growth values are typically greater than 1 after 1990. (An exception is 1991, when incomes declined slightly in the U.S.)

If you load in your own valuation functions and/or income growth adjustment factors, be sure that you have carefully considered the income year. For example, if your valuation functions are based on income in the year 2005, then the income growth adjustment should have a value of 1.0 in 2005, because no adjustment is necessary. As you forecast into the future, under the assumption that incomes go up over time, then your income growth adjustment factors would have values greater than 1.0 for years past 2005, and would have values less than 1.0 for years prior to 2005.

4.1.10.1 Add Income Growth Data

To add income growth data to BenMAP-CE, click on the **Manage** button below the **Income Growth Adjustments** box in the **Modify Datasets** window. The **Income Growth Adjustment Dataset Manager** window will appear.



In this window you may **Add** and **Delete** datasets. The section on the left under **Available Datasets** lists the **Income Growth Adjustments Datasets** that are currently in the setup database. The section on the right under **Income Growth Detail** presents the income growth adjustment factors in a selected dataset.

Click on the **Add** button. In the **Load Income Growth Adjustment Factors Dataset** window, type in the name of the dataset in the **Income Growth Adjustment Dataset Name** box, and then click on the **Browse** button to the right of the **Database** box to choose the dataset that you want to import and click **Open**. To confirm the file has proper formatting, click the **Validation** button before importing the file. In addition, you can edit the metadata for the imported file, to include references and descriptions, by clicking the **View Metadata** button.

Click **OK**. The **Income Growth Adjustment Dataset Manager** window will appear. Here you may view the data that you just loaded. From this window, you can also view or edit metadata by selecting an **Available Dataset** and **Income Growth Detail** and then clicking the **View Metadata** button.

4.1.10.2 Format for Income Growth Adjustment Data

Table 4-15a presents the variables that can be used in **Income Growth Adjustment Datasets** and Table 4-15b presents a sample of what a dataset might look like.

Table 4-15a. Income Growth Adjustment Dataset Variables

Variable	Type	Required	Notes
Year	Integer	Yes	The year of the data. Note that this will include historical estimates as well as forecasts.
Mean	Numeric (double)	Yes	Mean income growth adjustment factor.
EndPointGroup	Text	Yes	Endpoint group (e.g., <i>Chronic Asthma</i>).

Table 4-15b. Sample Income Growth Adjustment Dataset

Year	Mean	EndpointGroup
1990	1	Acute Bronchitis
1991	0.99788719	Acute Bronchitis
1992	1.0010463	Acute Bronchitis
1993	1.00313532	Acute Bronchitis
1994	1.00725281	Acute Bronchitis
1995	1.00950849	Acute Bronchitis
1996	1.01338995	Acute Bronchitis
1997	1.01823652	Acute Bronchitis
1998	1.02308881	Acute Bronchitis
1999	1.02832949	Acute Bronchitis
2000	1.03276563	Acute Bronchitis
2001	1.0327363	Acute Bronchitis
2002	1.03402805	Acute Bronchitis
2003	1.03695309	Acute Bronchitis
2004	1.04123259	Acute Bronchitis
2005	1.04486656	Acute Bronchitis
2006	1.04742563	Acute Bronchitis
2007	1.04867125	Acute Bronchitis
2008	1.04677248	Acute Bronchitis
2009	1.04110742	Acute Bronchitis
2010	1.04366136	Acute Bronchitis
2011	1.0449264	Acute Bronchitis
2012	1.04712892	Acute Bronchitis
2013	1.04852092	Acute Bronchitis
2014	1.05097616	Acute Bronchitis
2015	1.05361962	Acute Bronchitis
2016	1.05460298	Acute Bronchitis
2017	1.05676126	Acute Bronchitis
2018	1.0586977	Acute Bronchitis
2019	1.05993581	Acute Bronchitis
2020	1.06099141	Acute Bronchitis
2021	1.06246126	Acute Bronchitis
2022	1.06402493	Acute Bronchitis
2023	1.06561422	Acute Bronchitis
2024	1.06718934	Acute Bronchitis
2025	1.0687362	Acute Bronchitis
2026	1.07025254	Acute Bronchitis

4.2 Export and Import Setups

BenMAP-CE allows you to export and import entire databases (all **Available Setups**), individual setups (e.g., *United States*, *China*), and parts of individual setups (e.g. all **Grid Definitions**, or individual **Health Impact Function** datasets). This functionality can be used to archive data, share data with other BenMAP-CE users, as well as to allow you to move databases between computers or between versions of BenMAP (currently only supported for versions 1.3 and 1.4). In particular, all of the steps involved in creating a